

Miclat, Joana M,

BSCS-204

7OOP

Finals Lab Task 1. Encapsulation

```
class Car:
    # constructor
    def __init__(self, color:str, price: float, size: str):
        self.__color = color
        self.__price = price
        self.__size = size.upper()

    # getters
    def get_color(self):
        return self.__color

    def get_price(self):
        return self.__price

    def get_size(self):
        return self.__size

    # Setter
    def set_color(self, color: str):
        self.__color = color

    def set_price(self, price: float):
        self.__price = price

    def set_size(self, size: str):
        self.__size = size.upper()

    def __str__(self) -> str:
        size_desc = {'S': 'small', 'M': 'medium', 'L': 'large'}.get(self.__size, 'unknown')
        return f"Car ({self.__color}) - P{self.__price:.2f} - {size_desc}"
```

```
if __name__ == '__main__':

    print("Action: Invoking the car class car constructor using car('red', '19999.85', 'M')")
    car1 = Car(color="red", price=19999.85, size="M")
    print(car1)

    print("Action: Invoking the car Class car constructor using car('blue', '50000.00', 'L')")
    car2 = Car(color="blue", price=50000.00, size="L")
    print(car2)

    print("Action: Invoking the car Class car constructor using car('green', '12345.65', 's')")
    car3 = Car(color="green", price=12345.65, size="S")
    print(car3)
```

```
C:\Users\COMLAB\PycharmProjects\pythonProject1\venv\Scripts\python.exe C:\Users\COMLAB
Action: Invoking the car class car constructor using car('red', '19999.85', 'M')
Car (red) - P19999.85 - meduim
Action: Invoking the car Class car constructor using car('blue', '50000.00', 'L')
Car (blue) - P50000.00 - large
Action: Invoking the car Class car constructor using car('green', '12345.65', 's')
Car (green) - P12345.65 - small

Process finished with exit code 0
|
```